AIME 2026
24th International Conference on Artificial Intelligence in Medicine
OTTAWA, CANADA

# A Multimodal-based Framework for Smart ECG Report Generation

Tri-modal ECG signal, Computer Vision, and LLM for Automatic ECG Report Generation

Hong Duc Nguyen[1], Duc Tri Phan[2], S. Supraja[1]
[1] School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
[2] Early Mental Potential & Wellbeing Research (EMPOWER) Centre, Nanyang Technological University, Singapore

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Paper, refs & contact

## 1. Motivation

**Clinical Problem.** Electrocardiograms (ECGs) are essential for cardiac diagnosis, but raw recordings are often affected by movement artifacts, electrode misalignment, and electrical noise.

**Research Gap.** Existing ECG-text methods primarily rely on 1-D signal features and clinical text [1], overlooking the rich visual information in ECG images that clinicians use for interpretation [2].

## 2. Contributions

- A tri-modal framework integrating raw 12-lead ECG signals, visualized ECG images, and textual clinical reports.
- A unified feature sequence that supports both intra-domain and cross-domain attention.
- A lightweight adaptation strategy that reuses pretrained vision-language and language models (LVLMs and LLMs) instead of training a large model from scratch.

## 3. Experiment Settings

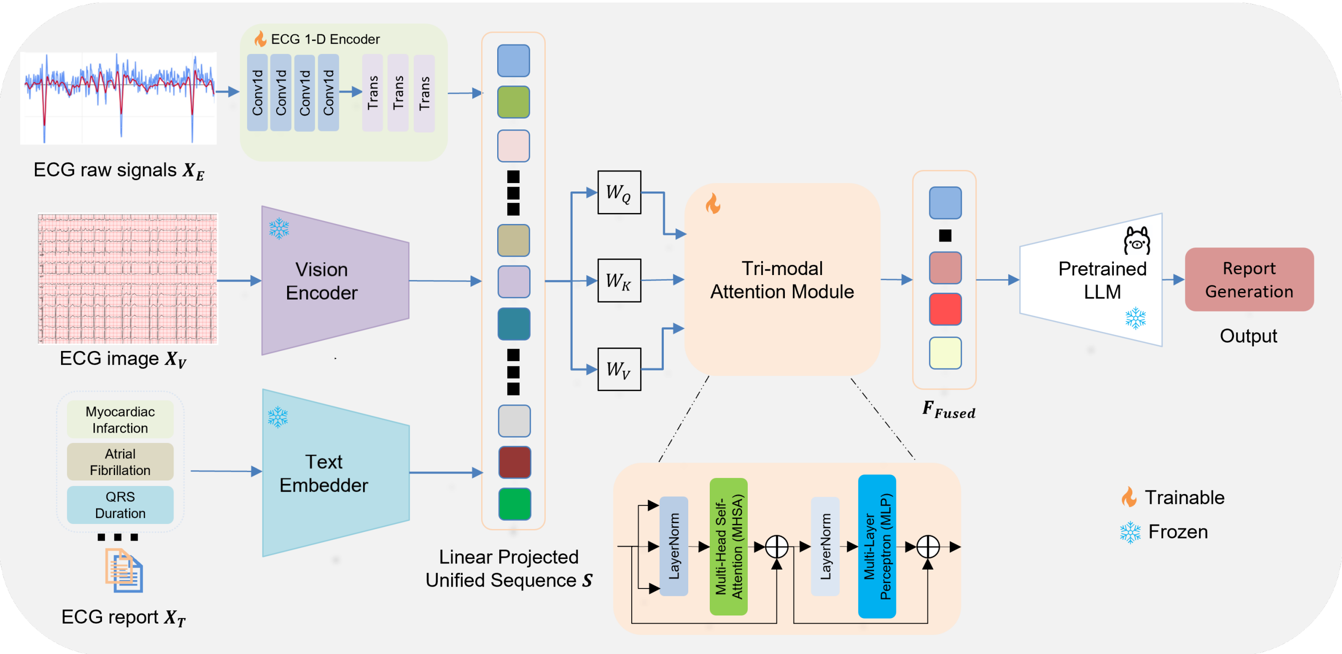
**Dataset** MIMIC-IV ECG with MEETI ECG images

**Metrics** BLEU-1/2/4 and ROUGE-L/1/2

**Training** 100 epochs with AdamW, cosine LR, early stopping

**Compared configurations.**
Vision encoders: LLaVA-Next, SigLIP, DINOv2-Giant, Biomed-CLIP.
LLM interpretation heads: LLaMA-3 and Mistral-Nemo.

## 4. Methodology




**Design Rationale.** The visual domain complements raw ECG signals and clinical text, allowing the model to reuse pretrained VLM/LLM representations while learning a compact multimodal alignment layer.

### ECG Encoder
1-D CNN captures local morphology; Transformer blocks capture temporal dependencies.

### Vision Encoder
Pretrained visual models encode grid patterns, waveform shapes, spikes, and visual structure.

### LLM Task Head
Fused multimodal features are mapped into the LLM space for ECG report generation.

### ECG Report Samples

"Upon examining the provided ECG image and computed measurements, several key features emerge that support the diagnosis. The heart rate is 84 bpm, indicating a sinus rhythm, which is consistent with the regular RR intervals observed. In Lead I, the QRS amplitude and duration are within normal limits, but the T wave amplitude is slightly elevated, suggesting a possible left axis deviation. Lead II shows a normal P wave amplitude and duration, with a PR interval that is within the normal range, indicating no significant conduction delay. The QRS complex in Lead III is normal, but the T wave amplitude is slightly elevated, which could be indicative of inferior wall changes. In Lead aVL, the QRS amplitude is normal, and the T wave amplitude is slightly elevated, which might suggest high lateral wall"

## 5. Results

**0.727** Best Mean Score

**0.773** Best ROUGE-L

**0.658** Best BLEU-4

**Table 1. Natural language generation results on MEETI.**

| VLM | LLM | B1 | B2 | B4 | RL | R1 | R2 |
|---|---|---|---|---|---|---|---|
| LLaVA | LLaMA-3 | 0.722 | 0.687 | 0.625 | 0.759 | 0.780 | 0.683 |
| LLaVA | Mistral | **0.745** | **0.701** | **0.658** | **0.773** | **0.786** | **0.696** |
| SigLIP | LLaMA-3 | 0.632 | 0.538 | 0.461 | 0.556 | 0.601 | 0.574 |
| SigLIP | Mistral | 0.648 | 0.572 | 0.491 | 0.583 | 0.628 | 0.602 |
| DINOv2 | LLaMA-3 | 0.651 | 0.582 | 0.502 | 0.637 | 0.673 | 0.601 |
| DINOv2 | Mistral | 0.682 | 0.639 | 0.579 | 0.710 | 0.701 | 0.673 |
| BioCLIP | LLaMA-3 | 0.629 | 0.535 | 0.479 | 0.548 | 0.593 | 0.565 |
| BioCLIP | Mistral | 0.636 | 0.535 | 0.449 | 0.573 | 0.612 | 0.593 |

**Discussion.** LLaVA-Next + Mistral-Nemo is the strongest combination overall across all reported metrics, suggesting that stronger visual representation and larger language generation capacity with high number of parameters help capture clinically relevant ECG morphology and report phrasing. DINOv2-Giant also benefits from Mistral-Nemo, indicating that the the structuralinterpretation head contributes meaningfully beyond the visual encoder alone. BiomedCLIP remains competitive with SigLIP despite significantly fewer parameters, showing the value of domain-specific visual pretraining for resource-constrained medical settings.

**Future Works.** We will extend the framework to ECG signal quality assessment and ECG denoising, and explore non-linear extracted ECG features to support future multimodal ECG assistants.

**Conclusion:** Aligning raw ECG signals, ECG images, and clinical reports enables stronger ECG report generation while reusing pretrained VLMs and LLMs with a compact fusion module.

**References:** [1] L. Simone, D. Bacciu, and V. Gervasi, "Quasi-orthogonal ECG-Frank XYZ transformation with energy-based models and clinical text," in *Proc. Int. Conf. Artificial Intelligence in Medicine*, 2024, pp. 249–253. [2] D. Zhang *et al.*, "MEETI: A multimodal ECG dataset from MIMIC-IV-ECG with signals, images, features and interpretations," *arXiv preprint arXiv:2507.15255*, 2025.